



Cambridge Lower Secondary Checkpoint

MATHEMATICS

0862/01

Paper 1

October 2024

MARK SCHEME

Maximum Mark: 50

Published

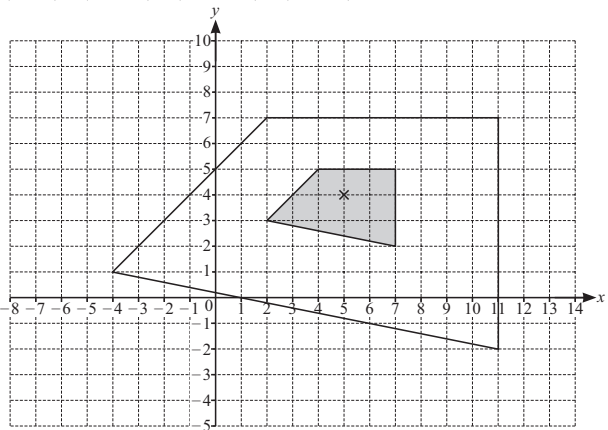
This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **10** pages.

Mark scheme annotations and abbreviations

FT	follow through after error
SC	special case mark
cao	correct answer only
dep	dependent
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
soi	seen or implied

Question	Answer	Marks	Part marks	Guidance
1	$7n^3$	1		Accept unsimplified, e.g. $(n^3) \times 7$
2	-3 and 2	2	Award 1 mark for each correct answer.	Answers must be in the correct order.
3	$60(^{\circ})$	2	Award 1 mark for 4×75 oe or 300	oe, e.g. repeated subtraction of 75 four times from any starting value (does not have to be from 360).
4	Quadrilateral drawn correctly at $(2, 7), (11, 7), (11, -2), (-4, 1)$ 	2	Award 1 mark for two or three vertices of the image correctly positioned or if the image has the correct size and shape but is wrongly positioned on the grid or for the correct centre of enlargement but wrong scale factor >1	Use template, mark intention. Must be on grid for this mark. Shape must fit on grid for 1 mark, i.e. with: SF 2 at $(3, 6), (9, 6), (9, 0), (-1, 2)$ SF 4 at $(1, 8), (13, 8), (13, -4), (-7, 0)$.
5	11	2	Award 1 mark for 2nd term = 4 or for $3^2 - 5 = 4$ oe seen or for $(\text{their previous term})^2 - 5$ correctly evaluated.	

Question	Answer	Marks	Part marks	Guidance
6	20 (km/h)	1		
7(a)	6	1		
7(b)	150 (cm ³)	1		
8	500 nm, 0.4 m, 70 cm	1		All three measurements in the correct order for the mark. Accept correctly converted units, e.g. 0.00005 cm, 40 cm (units must be present if they convert).
9	$\frac{1}{2} \times 22 \times \boxed{31} + \frac{1}{2} \times \pi \times \boxed{11}^2$	2	Award 1 mark for two correct values in the correct boxes.	For 11 accept $\frac{22}{2}$ or $22 \div 2$ i.e. consider the box acting like brackets. For $\frac{1}{2}$ accept equivalent fractions or 0.5

Question	Answer	Marks	Part marks	Guidance																																																																													
10(a)	<table><tr><th colspan="5">before</th><th></th><th colspan="5">after</th></tr><tr><td></td><td></td><td></td><td></td><td>9</td><td>4</td><td>(8)</td><td>(9)</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td>(2)</td><td>(1)</td><td>5</td><td>(1)</td><td>(5)</td><td>(8)</td><td>(8)</td><td>(8)</td></tr><tr><td></td><td>(5)</td><td>(5)</td><td>(5)</td><td>(4)</td><td>6</td><td>(2)</td><td>(4)</td><td>(7)</td><td>(7)</td><td>(8)</td></tr><tr><td>8</td><td>8</td><td>8</td><td>(8)</td><td>(3)</td><td>7</td><td>(5)</td><td>(6)</td><td>(7)</td><td>(8)</td><td></td></tr><tr><td></td><td>(9)</td><td>(7)</td><td>(6)</td><td>(5)</td><td>8</td><td>(2)</td><td>(5)</td><td></td><td></td><td></td></tr><tr><td></td><td>7</td><td>(5)</td><td>(4)</td><td>(3)</td><td>9</td><td>(1)</td><td>3</td><td></td><td></td><td></td></tr></table>	before						after									9	4	(8)	(9)							(2)	(1)	5	(1)	(5)	(8)	(8)	(8)		(5)	(5)	(5)	(4)	6	(2)	(4)	(7)	(7)	(8)	8	8	8	(8)	(3)	7	(5)	(6)	(7)	(8)			(9)	(7)	(6)	(5)	8	(2)	(5)					7	(5)	(4)	(3)	9	(1)	3				4	Award 1 mark for each of these correctly placed before: 9 as the lowest number in the 40s row - three 8s in the 70s row to make 78 the mode 7 as the highest number in the 90s row. after: 3 as the only extra value added in the correct place.	To a maximum of 2 out of 3 marks for an incorrect before side you can: - ignore other 9s in the 40 row - ignore other values in the table if the three 8s are correctly added provided 78 is still the mode, e.g. not another 5 in the 60s row - ignore another value less than or equal to 7 in the 90s row provided the numbers are correctly ordered.
before						after																																																																											
				9	4	(8)	(9)																																																																										
			(2)	(1)	5	(1)	(5)	(8)	(8)	(8)																																																																							
	(5)	(5)	(5)	(4)	6	(2)	(4)	(7)	(7)	(8)																																																																							
8	8	8	(8)	(3)	7	(5)	(6)	(7)	(8)																																																																								
	(9)	(7)	(6)	(5)	8	(2)	(5)																																																																										
	7	(5)	(4)	(3)	9	(1)	3																																																																										
10(b)	range	1		Accept any clear indication.																																																																													
11	12.65 (s) cao	1																																																																															
12	Not correct ticked and A correct explanation, e.g. - Rajiv has not followed the correct order of operations/BIDMAS/BODMAS - Rajiv multiplied 3 by 2 first - Rajiv should square first (then multiply by 2) - The correct answer is 18 - Rajiv has found $(2x)^2$.	1																																																																															

Question	Answer	Marks	Part marks	Guidance
13	True False True False ☒ ☐ ☐ ☒ True False True False ☒ ☐ ☐ ☒	2	Award 1 mark for three correct answers.	Accept any clear indication.
14	6	2	Award 1 mark for 32 [+ 10] or 42 seen.	Ignore cancelling of the 32 or 42, either can be in the numerator of a fraction. Do not accept if clearly spoilt, e.g. 32×2 , even done in stages, as this is p^6 .
15	$10 \leq l < 20$ $20 \leq l < 30$	1		Both intervals correct for the mark. Accept any exhaustive but unequal pair of intervals provided not overlapping, e.g. • $10 \leq l \leq 20$ and $20 < l < 30$ • $10 \leq l < 15$ and $15 \leq l < 30$

Question	Answer	Marks	Part marks	Guidance
16(a)	$\frac{9}{10}$ cao	2	Award 1 mark for equivalents to $\frac{9}{10}$ or for $[-]\frac{1}{10}$ oe	e.g. 0.9 , $\frac{63}{70}$ oe, e.g. $[-]\frac{7}{70}$, 0.1 1 mark can be awarded before any cancelling, converting or processing.
16(b)	$6\frac{3}{5}$ cao	3	Award 2 marks for equivalents to $6\frac{3}{5}$ or Award 1 mark for $\frac{8}{5}$ or $\frac{33}{8}$ If 0 or 1 scored, award 1 mark for correctly converting <i>their</i> improper fraction to a mixed number in its simplest form as their answer.	e.g. $\frac{33}{8}$, $\frac{264}{40}$, $6\frac{24}{40}$, 6.6 . 1 or 2 marks can be awarded before any cancelling, converting or processing.
17	$(x =)3$ and $(y =)5$	3	Award 2 marks for $(x =)3$ or $(y =)5$ or Award 1 mark for a correct method to eliminate one variable or for <i>their</i> two values satisfying one of the original equations.	Multiplying one or both equations by a constant to get equal coefficients and correct consistent addition or subtraction or making x or y the subject and substituting into the other equation.
18	Two angles where $0 < x < 80$ and $y = x + 100$	1		Both answers correct for the mark.

Question	Answer	Marks	Part marks	Guidance
19	Reflection and $x = -1$	2	Award 1 mark for reflection and Award 1 mark for $x = -1$ If 0 scored, award 1 mark for the correct rotation 90° clockwise of shape A about $(0, 0)$ seen.	Combinations of transformations score 0 marks (unless the diagram is correct or they are repeating the given transformation in the question). Treat extra properties as choice. Use template.
20		2	Award 1 mark for three or four from B, C, D, E and F correctly placed.	Accept calculations in place of letters. Look out for letters appearing in more than one region, mark the worst region if they do.

Question	Answer	Marks	Part marks	Guidance																								
21	True ☐ ☐ False ☒ ☒	1		Both answers correct for the mark. Accept any clear indication.																								
22(a)	50	1																										
22(b)	$\frac{2}{15}$ oe	2	Award 1 mark for $\frac{2}{5} \times \frac{1}{3}$ oe or a correct sample space diagram showing 15 outcomes with 2 indicated, e.g. by circling or $\frac{c}{15}$ where $c < 15$	For 2 marks, accept 0.13 or 13% or better (0.13). For 2 marks isw incorrect cancelling or conversion. You do not need to check every outcome when they list pairs to see if they are all correct but just make sure outcomes are not all incorrect (e.g. finding the sum of the two cards). Accept without outcomes, e.g. <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>2</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td><td>✓</td><td>✓</td></tr></table> For $\frac{c}{15}$ where $c < 15$ ignore method or sample space diagram. This can be awarded before any cancelling or converting.		1	2	3	4	5	2						3						4				✓	✓
	1	2	3	4	5																							
2																												
3																												
4				✓	✓																							

Question	Answer	Marks	Part marks	Guidance
23	6.7	3	Award 1 mark for $n = 2$ and Award 1 mark for $p = -4$ If 0 or 1 scored, award 1 mark for $0.067 \div 10^k$ correctly evaluated, k must be a negative integer for this mark.	Accept $\frac{67}{10}$, $6\frac{7}{10}$ for 3 marks. May be written next to n or p in the question. 2 marks may be implied by: 10^{2-4}, $2 - 4$, $[0.067 \div] 10^{-2}$, 0.067×10^2, $67 \times 10^{-3} \div 10^{-2}$, $0.067 \div 100 \times 10 000$ 1 mark may be implied by, e.g. 10^{2+k} Value of k must be shown or a calculation for k shown to award this mark.
24	$(c =) -24$	3	Award 2 marks for both expansions correct, e.g. $x^2 + 3x + 3x + 9$ or better and $x^2 - 3x + 5x - 15[+4x]$ or better or Award 1 mark for three correct terms out of one of these expansions $x^2 + 3x + 3x + 9$ or $x^2 - 3x + 5x - 15$	Or better, e.g. $x^2 + 6x + 9 +$ (accept all terms negative if they rearrange to make c the subject) $x^2 + 2x - 15$ $x^2 + 6x - 15$ $6^2 + 9 + c = 2x - 15 + 4x$ $9 + c = -15$ 6x implies two terms correct. 2x implies two terms correct or 6x if 4x added.